Appendix A — Methods description form (Track 2 methods)

Answers to the organisers' official methods description template (version with the required LLM question), reproduced here because the submission form has no separate upload field for it.

Team name

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Please describe your approach in detail (how you went from variant/mechanism to candidate medication(s)).

Allele-aware, patient-aware pipeline (scripts/t2/). (1) Mechanism model from Track 1: null allele p.(Leu737*) (UGA, NMD) + kinase-domain missense p.(Asn1002Lys) -> partial loss of function in a dosage-sensitive checkpoint. The CLASS of the missense allele (low abundance vs stable-but-dysfunctional) is unmeasured; every disease-associated substitution tested in or near the kinase domain was 5-10x less abundant and the known stable alleles lie outside that domain, so we carry both scenarios at a 0.75/0.25 prior and price the uncertainty into mechanistic fit, not evidence. AlphaFold structure: Asn1002 buried high-confidence residue near the MVA1 allele L1012P (destabilisation). (2) Knowledge base (hipotesis.yaml): 7 therapeutic lines + a safety card, each with PMID-cited evidence graded by level and direction (supporting/contrary): H1 stabilise BUBR1 (NAD+/SIRT2), H2 read-through/NMD inhibition of the nonsense allele, H3 mTORC1 (sarcopenia), H4 senolytics, H5 aneuploid-selective stress, H6 interferon/JAK-STAT (biomarker-gated), H7 proteostasis amplification/antioxidants (evaluated and declined). (3) Connectivity (cmap.py): consensus of 32 BUB1B CRISPR-knockout L1000 signatures (20 cell lines) via SigCom LINCS, two-sided rank enrichment against chemical perturbation signatures, cross-cell-line consistency, Enrichr interpretation (interferon/NF-kB/p53 up, E2F/G2M down). Used to check H6 for class consistency (JAK inhibitors reverse the signature), to support the safety card (DNA-damaging drugs mimic it) and to challenge H3 (mTOR inhibitors mimic it). (4) candidatos.py: seeded approved drugs (ChEMBL target expansion is available but added none in this run), annotated with ChEMBL and openFDA labels (paediatric use, boxed warnings) and penalised by the patient's HPO-derived risk profile; transparent score 0.30 evidence + 0.25 mechanistic fit + 0.20 paediatric feasibility + 0.25 patient-specific safety, with Dirichlet weight sensitivity; only approved medicines are ranked (discussion #5). (5) Clinical tiering by benefit-risk: tier 0 safety note plus rung-0 measurement; tier 1 intermittent dasatinib (highest-scoring approved drug), to be tested first in the BubR1 H/L1002P mouse; tier 2 an NAD+ precursor - niacin (nicotinamide ties on score but inhibits sirtuins) - gated on the allele class AND documented remission; tier 3 low-dose sirolimus for muscle with %MN-RET as a safety readout; tier 4 JAK inhibitors only if an interferon signature is measured; read-through closed on current allele-level evidence; proteostasis declined because raising HSF1 is a documented aneuploidy-TOLERATING event (Donnelly 2014). (6) nof1.py: an n-of-1 power simulation on micronucleated reticulocytes (%MN-RET), the only endpoint for this disease that moves in weeks. Validation ladder: rung 0 measure %MN-RET and its variability; patient cells (decides the allele scenario); iPSC myotubes; the BubR1 allelic mouse; then an n-of-1 protocol with pre-declared responder criteria.

(required) If commercially-available Generative AI was used, record the provider, the plan or tier, and the relevant setting in your methods description. A line is enough, for example, "Anthropic API, Claude Sonnet, commercial terms, no training on customer content."

Anthropic Claude via the Claude Code CLI (Claude Opus models), Claude Max (consumer tier), model-training opt-out enabled ('Help improve Claude' off), no feedback provided on any output, no content flagged for safety review - per the organisers' confirmed guidance of 28 Sep 2026 (discussion #2). Applicable terms: opted-out conversations are not used for training and are retained 30 days. Additional data minimisation: the model only saw code, aggregate counts, HPO terms and short candidate lists (<= 50 annotated variants); raw data and genome-scale derivatives were blocked from the agent by configuration. No patient variant was sent to any third-party API.

Was candidate identification automated (e.g., via a drug-target database query or LLM-assisted search), or primarily manual literature review and expert curation?

Hybrid: candidate expansion, annotation, safety flags, connectivity and scoring are automated; the hypotheses, evidence grading, mechanistic-fit values and final clinical tiering were curated manually (LLM-assisted literature search with every reference verified against PubMed).

Please describe any manual literature review or expert curation in detail.

PubMed searches at gene/pathway/drug level only (no patient data). Each claim in hipotesis.yaml cites a PMID whose title/abstract was retrieved and checked; contrary evidence is recorded explicitly (e.g., SIRT2 does not extend lifespan in wild-type mice; aneuploid-cell immune clearance argues against JAK inhibition). Mechanistic-fit values (0-1) are justified in the YAML. Automated label flags were reviewed manually.

Please state whether you relied solely on publicly available drug/target databases and literature (preferred), or whether proprietary data sources were also used.

Solely public databases and literature.

Please describe the public data sources used in detail (e.g., DrugBank, ChEMBL, ClinicalTrials.gov, OMIM, PubMed literature search).

PubMed/NCBI E-utilities; ChEMBL REST API (molecules, mechanisms, targets); openFDA drug labels; ClinicalTrials.gov API v2; LINCS L1000 via SigCom LINCS (l1000_xpr knockout and l1000_cp chemical signatures); Enrichr (MSigDB Hallmark 2020); UniProt O60566; AlphaFold DB AF-O60566-F1 v6; OMIM/HPO.

Please describe any proprietary data sources used in detail.

None.

How did you characterize the variant's mechanism (e.g., loss-of-function vs. gain-of-function, pathway disrupted, downstream biological consequence)?

Loss of function by reduced dosage: the nonsense allele is predicted to undergo NMD (no protein). For the kinase-domain missense we DO NOT assert a class: Suijkerbuijk 2010 splits MVA alleles into low-abundance and functionally-impaired classes, and although all disease-associated substitutions tested in or near the kinase domain were 5-10x less abundant, this residue has never been characterised, so scenario A (0.75) and scenario B (0.25) are both carried and a BUBR1 immunoblot in patient cells is the pre-declared gate. Pathway: spindle assembly checkpoint / mitotic checkpoint complex (APC/C-CDC20 inhibition) and kinetochore-microtubule attachment. Downstream: chromosome missegregation -> micronuclei -> cGAS-STING interferon/NF-kB and p53 responses (confirmed in the knockout signature), senescence/SASP, mTORC1 hyperactivity in muscle (mouse allelic series), tumour predisposition.

Please provide an estimate of time or effort spent on this analysis (if possible).

About 2 working days of analysis and writing; computation runs in minutes on a laptop (public APIs, cached).

Provide a method abstract (up to 500 words). Please include strengths and limitations of your approach as well as methodological detail.

We present an allele-aware, patient-aware repurposing pipeline for a child with biallelic BUB1B mosaic variegated aneuploidy (null p.Leu737* + kinase-domain p.Asn1002Lys). The mechanism is partial loss of function in a dosage-sensitive checkpoint; the missense allele class is unmeasured and both scenarios are carried. Three evidence streams were combined: (i) a PMID-cited knowledge base of seven therapeutic lines plus a safety card, scoring supporting and contrary evidence explicitly and weighting target validation below demonstrated benefit; (ii) connectivity analysis of 32 BUB1B-knockout L1000

signatures across 20 cell lines; (iii) automated annotation from ChEMBL and FDA labels, penalised by the patient's HPO-derived risk profile, with Dirichlet weight-sensitivity analysis. The knockout signature is interferon/NF-kB/p53-up and E2F/G2M-down; DNA-damaging chemotherapeutics reproduce it and mTOR inhibitors mimic it; JAK inhibitors reverse it with class consistency but at ranks 15-154 of 539 reversers, a targeted search rather than a discovery. Two evidence reviews changed most lines: read-through was closed, on current evidence, by the stop-codon context (UGA, unfavourable +4 A, NMD) plus evidence that ataluren's activity is a luciferase artefact; nicotinamide fell because rhabdomyosarcoma depends on NAMPT/NAD+ and nicotinamide inhibits sirtuins; senolysis became a preclinical experiment after the best senolytic RCT missed its endpoint; and sirolimus fell from first to tenth when its mouse support proved to come from monoallelic carriers. THE MAIN RESULT is a decision rule: classifying every line by what it does to a chromosomally unstable cell (correct fewer errors / clear the abnormal cells / tolerate help them cope) shows that no axis wins on evidence and that each corrective line is blocked by a different nameable fact about this child. Today the best-supported option is intermittent dasatinib (clear axis, narrowly ahead of the JAK inhibitor ruxolitinib); the mechanism-matched candidate, an NAD+ precursor (niacin), ranks eighth, and rises to first if the clinical team establishes the allele class and documented remission - so the first recommendation is a measurement, not a prescription. In a cancer-predisposition syndrome a premalignant clone is an abnormal cell too. Deliverables: a two-page pharmacogenomic safety card, usable now; %MN-RET, a blood measurement never made in MVA, diagnostic first and then an endpoint with its power calculation; a ranked and conditioned candidate list, and a four-rung validation ladder (patient cells, iPSC myotubes, the published BubR1 allelic mouse model, then an n-of-1 protocol with pre-declared responder criteria) built on verified precedents (lonafarnib, sirolimus in Castleman disease). Strengths: per-allele strategies, explicit contrary evidence that demoted our own candidates, patient-specific safety beyond drug labels, and reproducible gene-agnostic code. Limitations: efficacy evidence is preclinical; connectivity signatures come from tumour cell lines after acute knockout; mechanistic-fit values are expert judgements and the ranking is sensitive to two of them, so the counterfactual is published rather than a robustness claim; every ranked drug was hand-seeded (automated target expansion added none); the %MN-RET baseline and variability used in the power calculation have never been measured in MVA; regulatory routes are orientation, not legal advice. An independent blind adversarial review of an earlier draft drove these corrections.